

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456250.6094 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.025 +/- 0.055
Transit depth (Rp/Rs)^2	0.06 +/- 0.27 [%]
Orbital Inclination (inc)	83.71 +/- 1.91
Airmass coefficient 1 (a1)	0.986 +/- 0.032
Airmass coefficient 2 (a2)	0.02 +/- 0.18
Scatter in the residuals of the lightcurve fit is	3.31 %
Best Comparison Star	#3 - [398, 165]
Optimal Aperture	2.74
Optimal Annulus	9.0
Transit Duration (day)	0.014 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.025 ± 0.055 vs 0.1592 (deviation 1.44σ)
Duration fit vs published	0.014 d vs 0.0946 d (deviation 2.17σ)
Mid-time O–C	2.4 min
Depth SNR	0.3
Residual scatter	3.31 %

Light curve

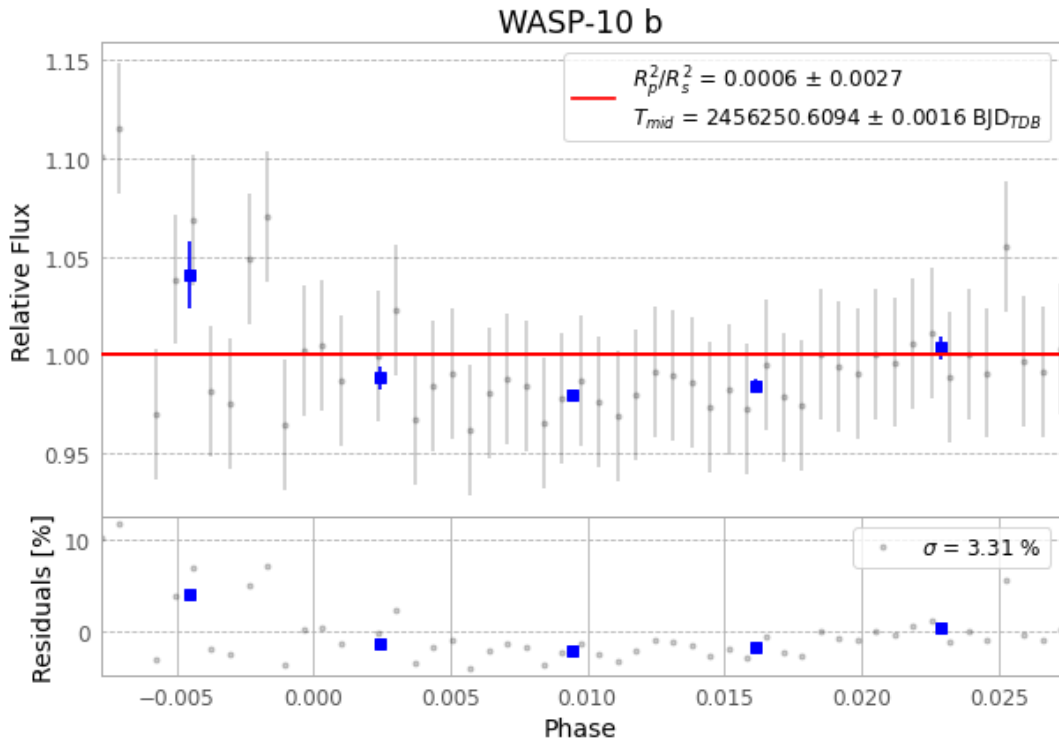


Figure 1 — FinalLightCurve_WASP-10 b_2012-11-18.png (SHA-256 e448843bc39af887...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [402, 263], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dcee0cc4fbd05938...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

53 FITS frames (35 MB), WASP-10121119015712.FITS → WASP-10121119043312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-121119.log (74749a3ec8a7...), FinalLightCurve_WASP-10 b_2012-11-18.png (e448843bc39a...), FinalLightCurve_WASP-10 b_2012-11-18.csv (7598dab517b3...)

Compute resources

Wall time 131.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 00:40Z.